Operations Management - II
Case: Moto Tech Manufacturing Company Case Analysis

NAMAN JAIN

ROLL NO - FT272022

SECTION 2

COURSE – OPERATIONS MANAGEMENT - II

Q1 - What immediate business problem triggered John Tagole to seek help from The Famous Five? Why is the issue strategically important to MotoTech?

The trigger is a phone call to John Tagole, MotoTech Manufacturing's (MM's) founder and CEO, from Danika Katz, VP of Procurement at Semicon International, one of MM's largest customers of six years. Semicon's engineers have drastically tightened the design specification for a component going into a new line of high-definition, 3D-imagery televisions. MM's diffusion process comfortably meets Semicon's old specification but does not meet the new, narrower one (still 2,900 to 3,100 Å around a 3,000 Å target, but now enforced far more strictly), and Danika warns that unless MM can meet it, Semicon may take its business elsewhere. Tagole calls in his old MBA classmates, "The Famous Five," now running their own consulting firm, because this is not a problem he can solve by simply working harder on the line: the process itself may not be capable of the tighter tolerance, no matter how carefully any single batch is run.

Strategically, this matters for MM well beyond this one account. Tagole has already been troubled by MM's lower yield and higher rejection rate compared with competitors, and overseas rivals from Taiwan, Japan, Singapore, Korea, and China have been encroaching on MM's traditional customer base. Losing Semicon, one of MM's largest customers, at what the case calls "a critical juncture" in MM's 6-year history would be a serious setback to Tagole's ambition of growing MM into a global chip supplier. And because Semicon's move reflects where the wider industry is headed (tighter tolerances driven by more demanding end products), a supplier that cannot demonstrate control and capability now will keep losing this fight as specifications keep tightening industry-wide.

Q2 - Which stage of the manufacturing process is the focus of the case? What quality characteristics are controlled, and what are the target and customer specification limits?

The focus is the diffusion stage of MM's wafer fabrication process, the step where a layer of silicon dioxide is deposited onto each wafer inside a diffusion furnace (as many as 200 wafers batch-processed together). The controlled quality characteristic is the thickness of that deposited layer, measured in Angstroms (Å), a variable (continuously measured) characteristic, which is why the case uses X-bar and R charts rather than p- or c-charts.

Target and customer specification limits (Part E3 of the template): Target = 3,000 Å; Lower Specification Limit (LSL) = 2,900 Å; Upper Specification Limit (USL) = 3,100 Å. These are Semicon's requirements, not internal process limits, and they define the ±100 Å window the process must fit inside.

Q3 - Describe the sampling plan used for the control-chart analysis: batch size, subgroup/sample size, and number of batches observed.

Batch size: 200 wafers per production batch

Subgroup (sample) size: 3 wafers, drawn from each batch

Number of batches observed: 72 subgroups for the pre-improvement period and 72 subgroups for the post-improvement period, giving 216 wafers measured in each period.

This is a small, frequent-sampling plan, which is typical for variable-data control charts: small subgroups (n = 3) taken often are more sensitive to a shift in the process mean than large subgroups taken rarely.

Q4 - In the context of MotoTech, distinguish between a process being statistically “in control” and a process being “capable.”

A process is statistically "in control" when the variation it shows is only common-cause (random) variation, with no points or patterns outside the control limits on the X-bar and R charts. Being in control says nothing about whether the output meets the customer's specification; it only says the process is behaving predictably, the same way, batch after batch.

A process is "capable" when its natural variation, typically ±3 sigma around the mean, i.e., 6 sigma total width fits inside the customer's specification width (USL − LSL), and is centered well enough that neither tail crosses a spec limit. For MotoTech, the post-improvement process turns out to be a clean illustration of the distinction: Part C shows it is in control, but Part E3 shows it is still not capable of Semicon's tightened specification. A process can be perfectly predictable and still be the wrong process for the job.

Q5 - Why should the R chart be examined before drawing a conclusion from the X-bar chart?

The X-bar chart's control limits are calculated from R-bar (UCL = X-double-bar + A2 x R-bar; LCL = X-double-bar − A2 x R-bar). If the R chart itself is out of control, R-bar is not a reliable estimate of the process's true, common-cause variation; it is inflated or distorted by whatever special-cause event produced the erratic ranges. Using a contaminated R-bar to build the X-bar limits produces X-bar limits that are themselves untrustworthy: they might be too wide or, less commonly, positioned incorrectly. The R chart has to be confirmed in control first, because it validates the very control limits the X-bar chart depends on.

Part A: Strategic Response to the Customer Specification Change

Q.A1 - What could happen if MotoTech delays action? (Consider customer relationship, production continuity and immediate competitive pressure.)

In the short term, MM keeps shipping wafers produced by a process that, as Part E3 shows, is not capable of the new specification even when it is behaving predictably. That means a steady trickle of out-of-spec wafers reaching Semicon regardless of how carefully any single batch is run, since the gap is built into the process's own variation and centering, not into isolated mistakes. That erodes trust quickly: Semicon will see a supplier whose defect rate does not visibly improve over time, and production continuity is at risk if Semicon starts rejecting lots, holding shipments for extra inspection, or dropping MM outright, as Danika Katz has already warned. Competitively, any delay is also time overseas rivals from Taiwan, Japan, Singapore, Korea, and China, already encroaching on MM's traditional customer base, can use to demonstrate a capable process and pick up the business MM is at risk of losing.

Q.A2 - What are the risks of assuming customer specifications will relax? (Consider capability development, competition and strategic positioning.)

Tagole's own hope, that the new limits might be "temporary, imposed by a young, overenthusiastic design engineer," is a guess he has no real evidence for. Betting on it assumes MM has leverage it may not have: a tightened specification tied to a new, higher-end 3D-imagery television line almost certainly reflects Semicon's own downstream engineering requirements, not an arbitrary preference, so it is unlikely to move back. Waiting also means MM does not invest in the process-capability improvements it will eventually need anyway, whether for Semicon or for any other customer who tightens tolerances next, at a time when global competition is already squeezing MM's yield advantage. Meanwhile, competitors are not waiting: a supplier that treats a customer's specification as fixed and adapts its own process is the one that wins the next contract renewal. Positioning MM as reactive, rather than as a supplier who solves precision problems, is itself a long-term strategic cost.

Q.A3 - State the action you recommend to John Tagole and justify it in 2-3 evidence-based points.

MotoTech should not wait for Semicon to relax the specification. It should treat the tightened spec as the new baseline and invest immediately in closing the capability gap. 3 evidence-based points support this:

1. The post-improvement data already proves the process can be brought into statistical control (Part C); MotoTech is not starting from 0, it is extending work already underway.

2. Part E3 shows Cpk (0.32) is far below Cp (0.83), and both are below the conventional capability benchmark of 1.33; the gap between Cp and Cpk points to a correctable centering problem, the mean sits too close to the USL as well as a variation problem, so there is a concrete, prioritized fix available rather than an open-ended one.

3. Acting now protects the relationship and buys credibility with Semicon during the fix, whereas waiting risks the account before any improvement is shown at all.

Part B: Pre-Improvement Process Control Analysis

Q.B1 - Construct the X-bar and R control charts. Show your calculations and insert both charts.

For each of the 72 subgroups (n = 3 wafers), the sample mean and sample range were calculated:

Sample mean: $\bar{X}_i = (X_1 + X_2 + X_3) / 3$

Sample range: $R_i = \max(X_1,X_2,X_3) - \min(X_1,X_2,X_3)$

The grand mean and average range across all 72 subgroups (all 72 rows are in the accompanying Excel file, sheet "Pre-Improvement"):

$\bar{\bar{X}}$ = average of the 72 sample means = 3.1326 (thousand Å)

R-bar = average of the 72 sample ranges = 0.1165 (thousand Å)

Using the three-sigma control chart factors for a subgroup size of n = 3 (A2 = 1.02, D3 = 0, D4 = 2.57):

UCL (X-bar) = X-double-bar + A2 x R-bar = 3.1326 + 1.02 x 0.1165 = 3.2514

LCL (X-bar) = X-double-bar − A2 x R-bar = 3.1326 − 1.02 x 0.1165 = 3.0138

UCL (R) = D4 x R-bar = 2.57 x 0.1165 = 0.2994

LCL (R) = D3 x R-bar = 0 x 0.1165 = 0.0000

Q.B2 - Use the R chart to determine whether process variation is in statistical control. Identify the evidence (sample numbers and/or non-random patterns).

No. Two subgroups exceed the R chart's UCL of 0.2994: sample 8 (R = 0.328) and sample 40 (R = 0.589). Sample 40 is a particularly large spike, nearly twice the UCL, indicating a special-cause event that made the spread within that subgroup unusually wide, one wafer measured far from the other 2. The R chart is not in control.

Q.B3 - Use the X-bar chart to determine whether the process mean is in statistical control. Identify the evidence.

No. 10 subgroups fall outside the X-bar chart's control limits (3.0138 to 3.2514): samples 5, 27, 32, 33, 40, 41, 61, 64, 70, and 72. Sample 40 is again the most extreme point, with a sample mean of 3.675, driven by the same special cause that inflated its range. The X-bar chart is not in control either.

Q.B4 - If the R chart is out of control, are the X-bar chart control limits valid for making a final judgment about the mean? Explain why or why not.

No, not fully. Because R-bar feeds directly into both the UCL and LCL formulas for the X-bar chart, an out-of-control R chart means R-bar is contaminated by the same special causes that are inflating individual ranges. The X-bar control limits calculated from that R-bar are therefore not a trustworthy benchmark for judging the mean. In practice, the correct sequence is to investigate and remove the special-cause points from the R chart first, recompute R-bar and the X-bar limits from the remaining, in-control data, and only then re-evaluate whether the mean is in control. The ten X-bar violations reported in B3 are real signals that something is wrong, but the specific limits used to detect them would need to be recalculated once the R chart's assignable causes are addressed.

Part C: Post-Improvement Process Control Analysis

Q.C1 - Construct the X-bar and R control charts for the post-improvement process.

Using the same formulas as Part B1, applied to the 72 post-improvement subgroups (full data and row-by-row calculations in the Excel file, sheet "Post-Improvement"):

X-double-bar = 3.0613 (thousand Å)

R-bar = 0.0677 (thousand Å)

UCL (X-bar) = 3.0613 + 1.02 x 0.0677 = 3.1303

LCL (X-bar) = 3.0613 − 1.02 x 0.0677 = 2.9923

UCL (R) = 2.57 x 0.0677 = 0.1739

LCL (R) = 0 x 0.0677 = 0.0000

Q.C2 - Use the R chart to determine whether the post-improvement process variation is in statistical control. Provide evidence

Yes. All 72 sample ranges fall between 0.0000 and 0.1739; there are no points outside the control limits and no obvious non-random pattern (run, trend, or cycle) in the R chart. Process variation is stable after the intervention.

Q.C3 - Use the X-bar chart to determine whether the post-improvement process mean is in statistical control. Provide evidence.

Yes. All 72 sample means fall between 2.9923 and 3.1303, with no points outside the limits. Because the R chart is confirmed in control first (C2), these X-bar limits can be trusted, and the conclusion that the mean is in control is a valid one.

Part D: Recommendation Based on Parts B and C

The pre- and post-improvement charts tell a clear before-and-after story. Before the intervention, both the R chart and the X-bar chart showed real, assignable-cause variation: 2 range violations and 10 mean violations, including 1 dramatic spike that likely reflects a specific, identifiable event, like a measurement error, an equipment glitch, or a process upset rather than routine variation. After the intervention, both charts are completely clean: no points outside the limits on either chart, and R-bar itself dropped by roughly 42% (from 0.1165 to 0.0677), meaning wafer-to-wafer consistency within a batch also improved substantially, not just the batch-to-batch average.

The Famous Five traced the assignable-cause variation to a specific, identifiable event: the process went out of control whenever the plant's air conditioning system was shut down for preventive maintenance, letting the diffusion room's temperature drift. Their fix was to install a back-up air conditioner and hold the diffusion room at a steady 60°F; until the new unit could be installed, they had preventive maintenance rescheduled to weekend nights, when the diffusion process itself was stopped. MotoTech should standardize exactly this fix, since it is demonstrably responsible for taking the process from out-of-control to in-control. It fits the data well too: sample 40's dramatic spike pre-improvement, a mean of 3.675, nearly 0.5 above every neighboring batch is exactly the signature of a batch run during an uncontrolled temperature swing, not routine noise, which is why it deserved root-cause investigation rather than being averaged away.

What management should investigate next, based only on what Parts B and C show: the post-improvement process, while now stable, still runs with a mean (3.0613) that sits noticeably above the 3.000 target and close to the tightened USL of 3.100. Parts B and C only establish control, not capability, but the post-improvement X-bar chart already hints that centering, not just variation, may need further attention, which is exactly what Part E3 goes on to confirm.

Part E3: Process Capability Cp and Cpk (Post-Improvement Process Only)

Q.E3(a) - Compute Cp and Cpk. Show all steps.

Given: USL = 3.100, LSL = 2.900, Target = 3.000 (thousand Å); from Part C, R-bar = 0.0677 and X-double-bar = 3.0613 (post-improvement).

Step 1: Estimate process sigma from R-bar, using the standard control-chart factor d2 for a subgroup size of n = 3 (d2 = 1.693):

sigma = R-bar / d2 = 0.0677 / 1.693 = 0.0400

Step 2: Cp (precision relative to the specification width, ignoring centering):

Cp = (USL − LSL) / (6 x sigma) = (3.100 − 2.900) / (6 x 0.0400) = 0.200 / 0.240 = 0.834

Step 3: Cpu, Cpl, and Cpk (bringing the actual process mean into the calculation):

Cpu = (USL − X-double-bar) / (3 x sigma) = (3.100 − 3.0613) / (3 x 0.0400) = 0.323

Cpl = (X-double-bar − LSL) / (3 x sigma) = (3.0613 − 2.900) / (3 x 0.0400) = 1.345

Cpk = MIN(Cpu, Cpl) = MIN(0.323, 1.345) = 0.323

Q.E3(b) - Is the post-improvement process capable of meeting Semicon’s specification? State the capability benchmark you are using and justify your conclusion.

No. Using the common industry benchmark of Cpk >= 1.33, a widely used threshold, sometimes relaxed to 1.0 for less critical characteristics as the capability benchmark, MotoTech's post-improvement Cpk of 0.323 falls far short. Even the more forgiving Cp of 0.834 is below 1.0, meaning the process's natural 6-sigma spread (about 0.240 thousand Å) does not even fit inside Semicon's 0.200 thousand Å specification window on its own, before centering is considered at all. The post-improvement process is in statistical control, but it is not capable of meeting Semicon's tightened specification.

Q.E3(c) - Compare Cp and Cpk. What does the gap tell you about process precision and centering?

Cp (0.834) measures only whether the process's spread would fit the specification window if it were perfectly centered on the target; it says nothing about where the process actually sits. Cpk (0.323) measures the worst-case distance from the mean to the nearer specification limit, and is always less than or equal to Cp. Here the gap between them is large: Cpk is only 39% of Cp. That gap is entirely attributable to poor centering, not poor precision. The process mean (3.0613) sits noticeably above the 3.000 target, close enough to the USL (3.100) that Cpu (0.323) is far worse than Cpl (1.345). If the process were simply re-centered on 3.000 without changing its variation at all, Cpk would rise all the way to Cp's value of 0.834, more than double its current level, though still short of full capability.

Q.E3(d) - Is Cp alone an appropriate metric for this process? Explain why or why not and state the managerial implication of Cpk.

No, Cp alone is not appropriate for MotoTech's process. Cp assumes the process is centered on the target, and this one clearly is not (a 0.0613 thousand-Å offset from a 3.000 target, on a specification half-width of only 0.100). Reporting Cp alone (0.834) would understate the real risk: it would suggest the process is only modestly short of capable, when in fact wafers are running much closer to the upper specification limit than the raw Cp number implies. The managerial implication of Cpk is that it forces attention onto both levers available to fix the problem, precision (reduce variation, i.e., lower sigma) and centering (shift the mean back toward 3,000), and in this case it shows clearly that centering is the more urgent and more immediately actionable of the 2: it requires adjusting a process setpoint, not necessarily a harder engineering investment in reducing variation. Reporting Cpk, not Cp, should be the standard practice MotoTech adopts for any specification-sensitive characteristic going forward.